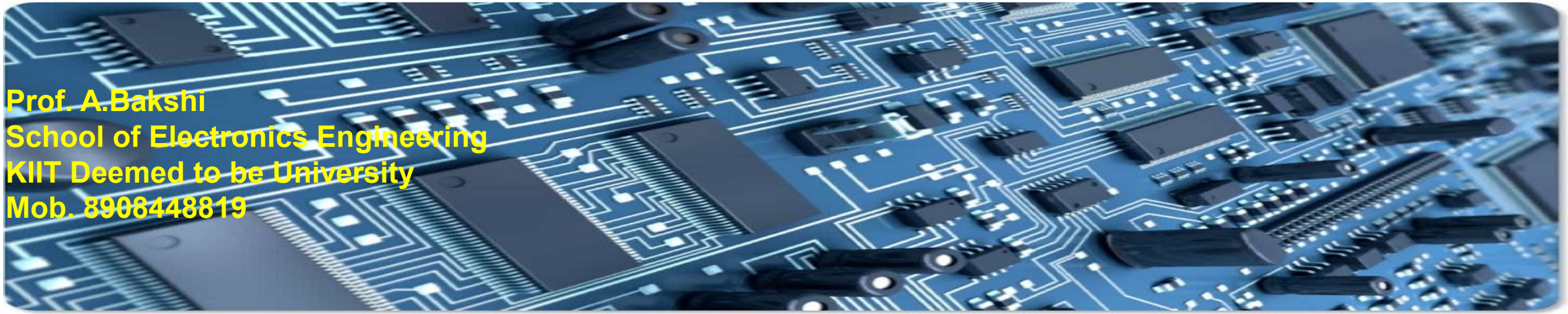
Basic VLSI System Design



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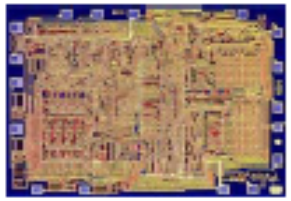
Introduction to VLSI Design

Some History

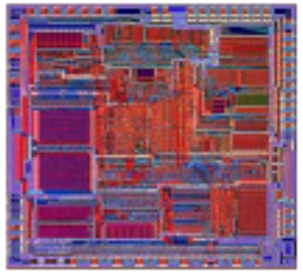
Invention of the transistor (BJT) 1947
Shockley, Bardeen, Brattain – Bell Labs

Single-transistor integrated circuit 1958
Jack Kilby – Texas Instruments

Invention of CMOS logic gates 1963
Wanlass & Sah – Fairchild Semiconductor



First microprocessor (Intel 4004) 1970
2,300 MOS transistors, 740 kHz clock frequency



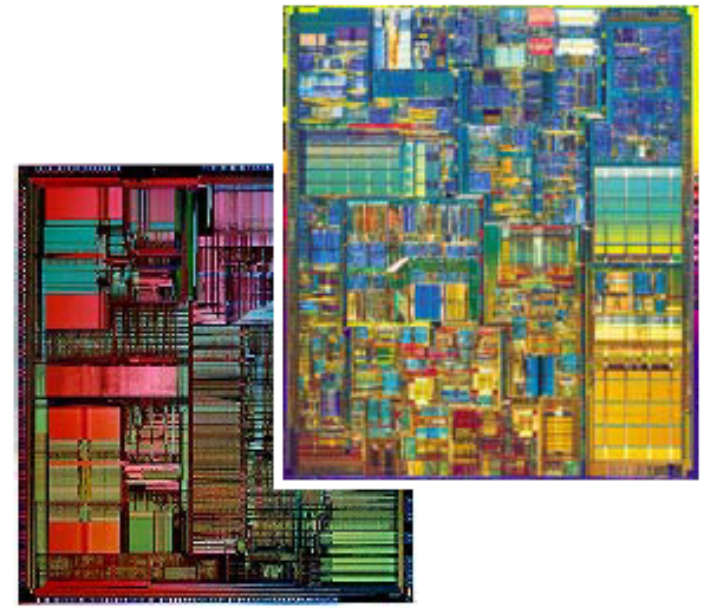
Very **L**arge **S**cale **I**ntegration 1978
Chips with more than ~20,000 devices

More Recently

Ultra **L**arge **S**cale **I**ntegration

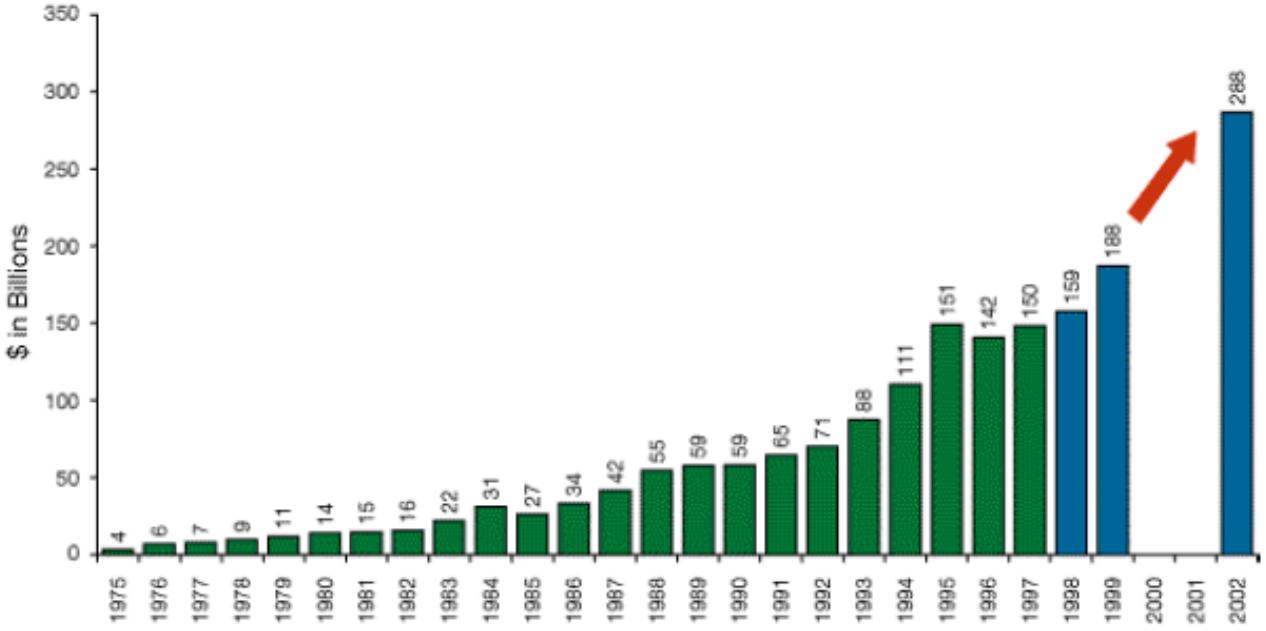
System **o**n **C**hip (**SoC**)

20 ~ 30 million transistors in 2002



The chip complexity has increased by a factor of 1000 since its first introduction, but the term **VLSI** remained virtually universal to denote digital integrated systems with high complexity.

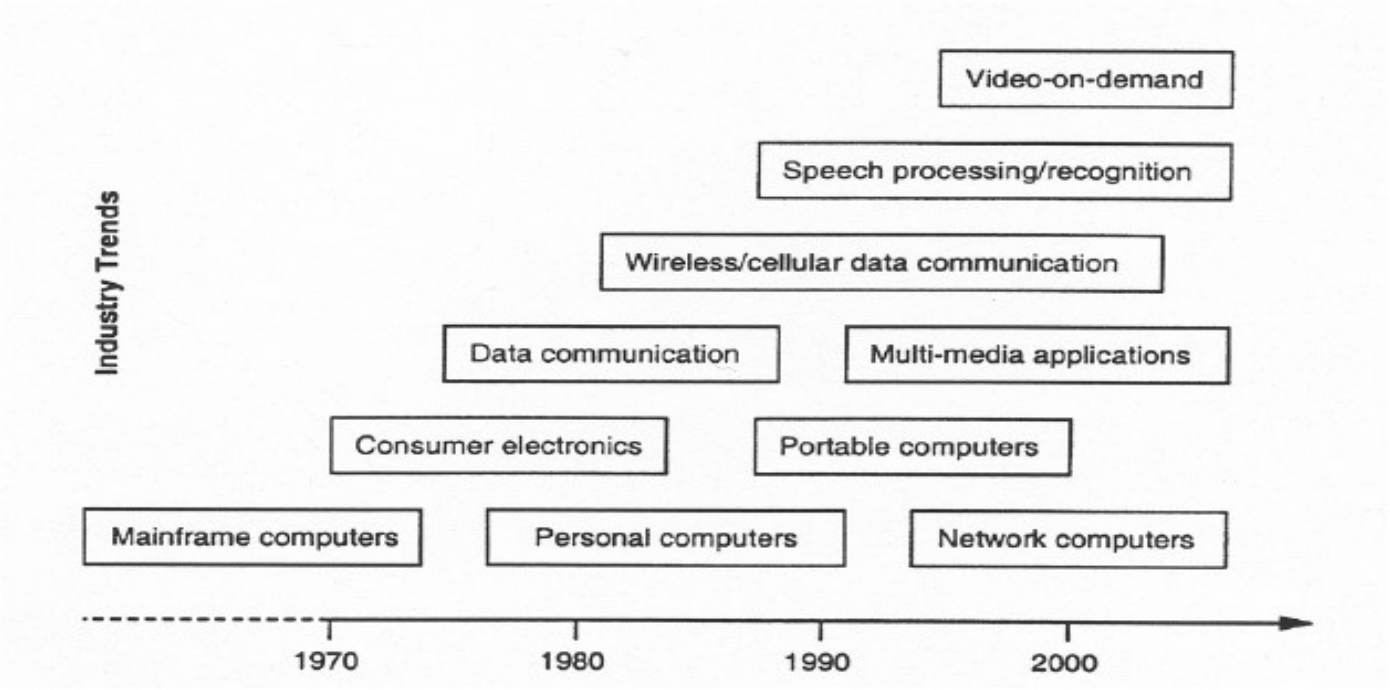
Economic Impact



Source: Dataquest, April 28, 1998

As a result of the continuously increasing integration density and decreasing unit costs, the semiconductor industry has been one of the fastest growing sectors in the worldwide economy.

Industry Trends



Large
Centralized
Expensive

Small / Portable
Distributed
Inexpensive

The electronics industry has achieved a phenomenal growth over the last two decades, mainly due to the rapid advances in integration technologies, large-scale systems design - in short, due to the advent of VLSI.

Industry Trends

High performance
Low power dissipation
Wireless capability
etc…

More portable, wearable, and more powerful devices for ubiquitous and pervasive computing…

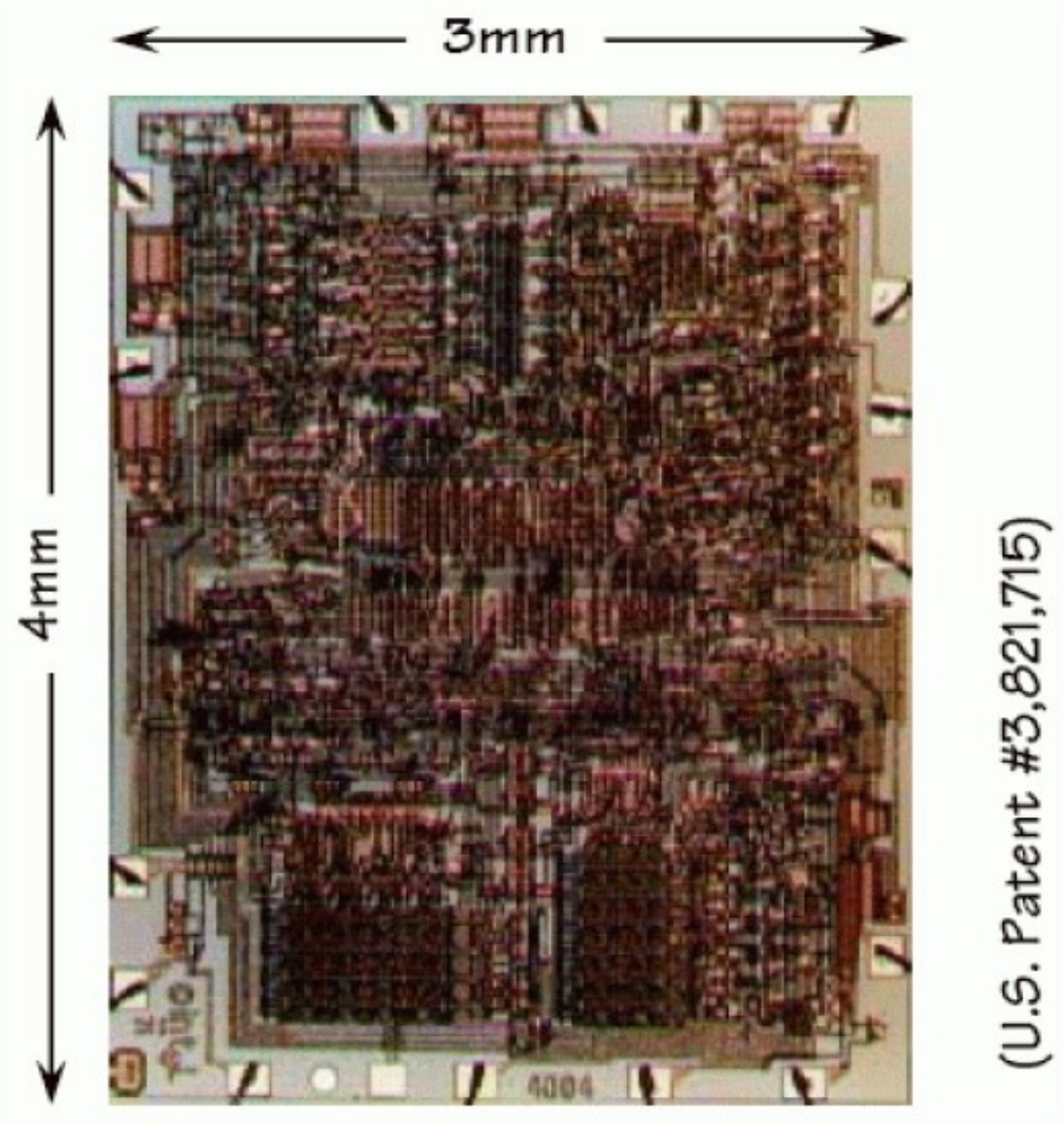
INTegrated ELectronics=Intel (1968)

[2] Gordon Moore [right] relaxes with fellow pioneers of the electronic age: Robert Noyce [center] and Andrew Grove [left]. Moore and Noyce contributed to the development of the planar IC. Grove is now president and chief executive officer of Intel Corp.

- Noyce and Moore leave Fairchild and found Intel. No business plan, just a promise to specialize in memory chips. They and Art Rock raise $2.5M in two days and move to Santa Clara. By 1971 Intel had 500 employees; by 1983 it had 21,500 employees and $1.1B in sales.

- Pioneers of the Electronic Age

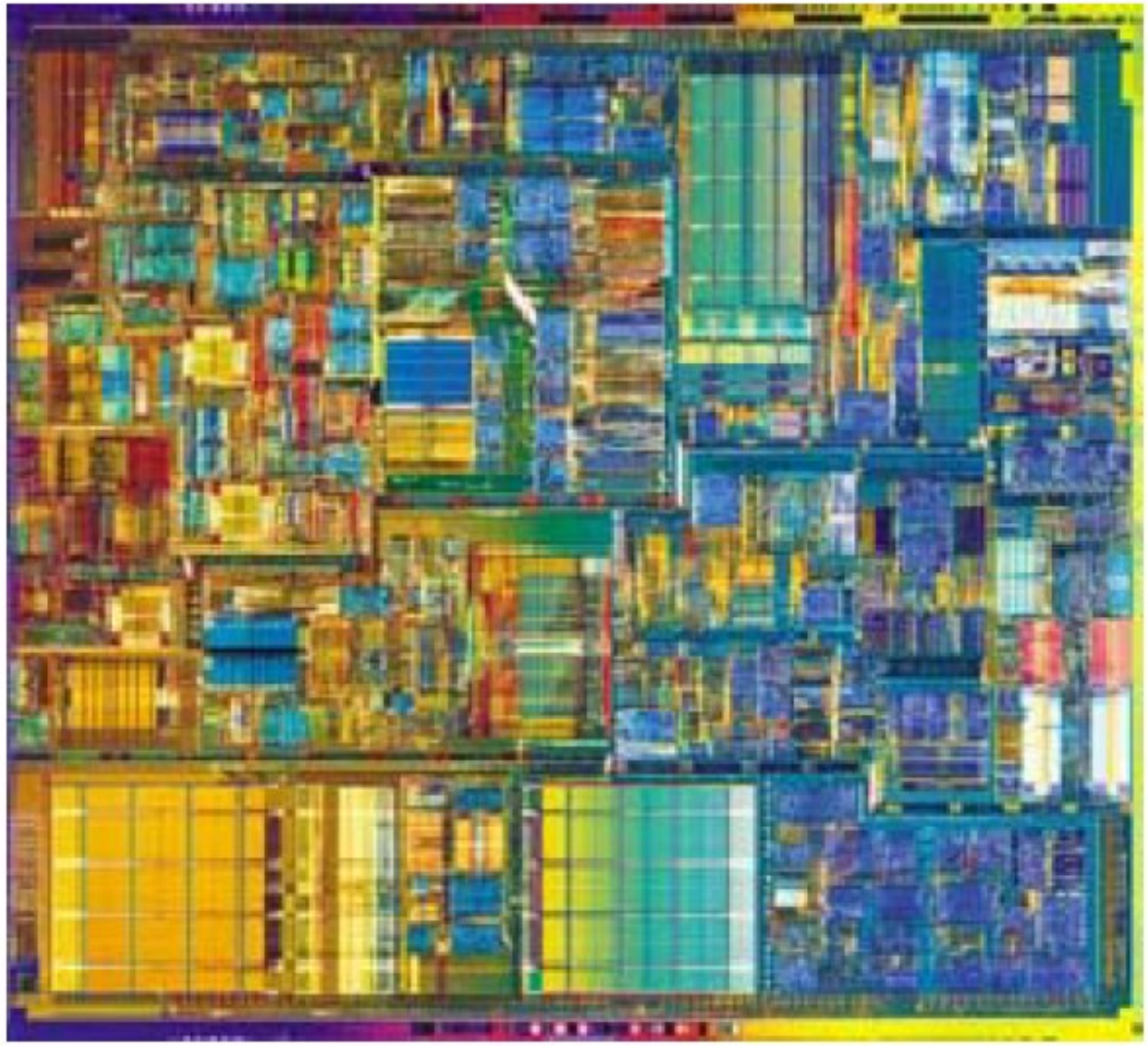
Intel 4004 Micro-Processor (1971)



(U.S. Patent #3,821,715)

- The 4004 had 4-bit buses and a clock rate of 1M Hz. It had 1000 transistors and was built in a 10 um process.
- In 1971 Intel introduces the first microprocessor, designed by Ted Hoff. It never captured much interest in the market and was soon eclipsed by its more capable brethren.

Intel Pentium (IV) microprocessor



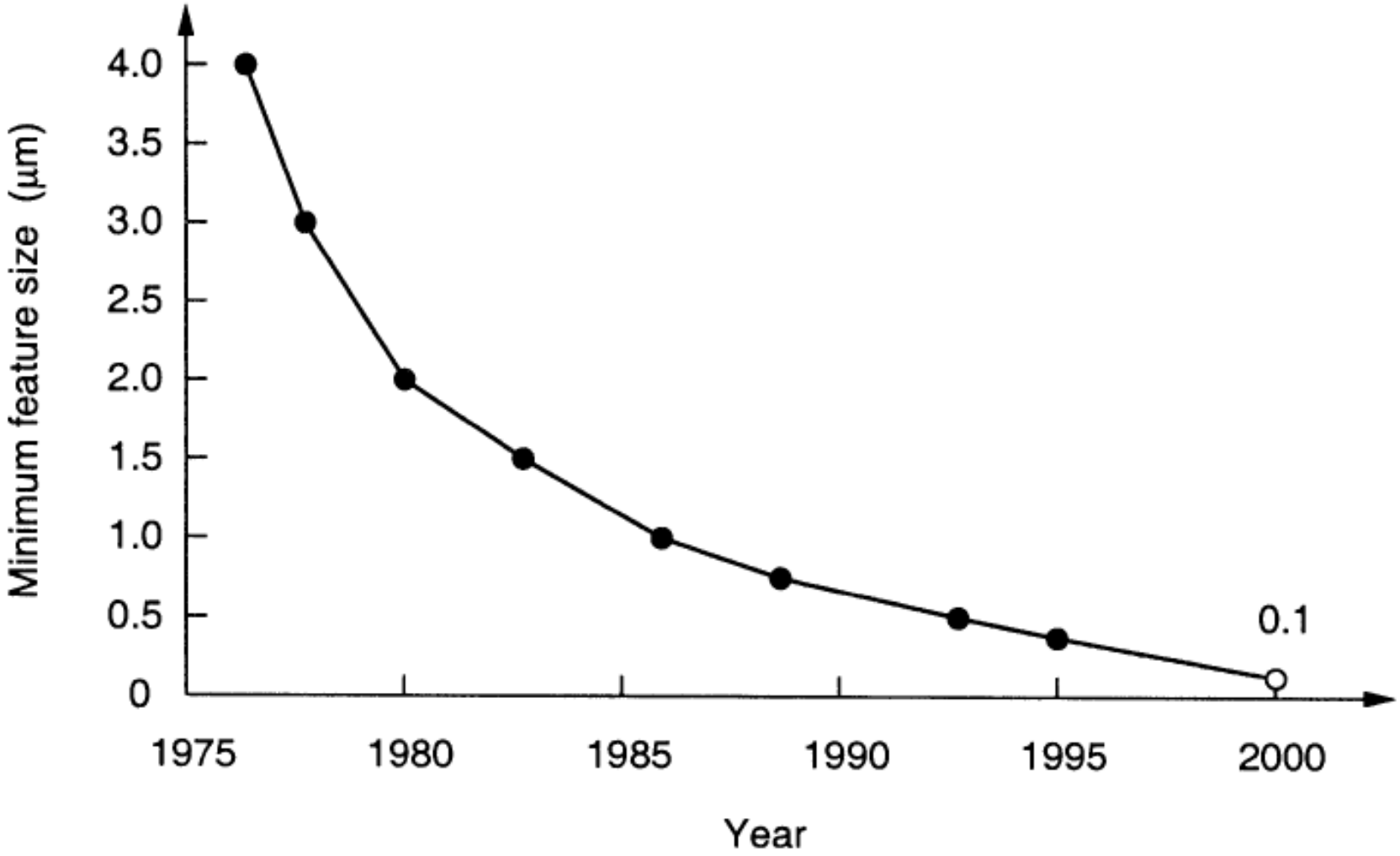
0.13μ process

55 million transistors

2.4GHz clock

145mm^2

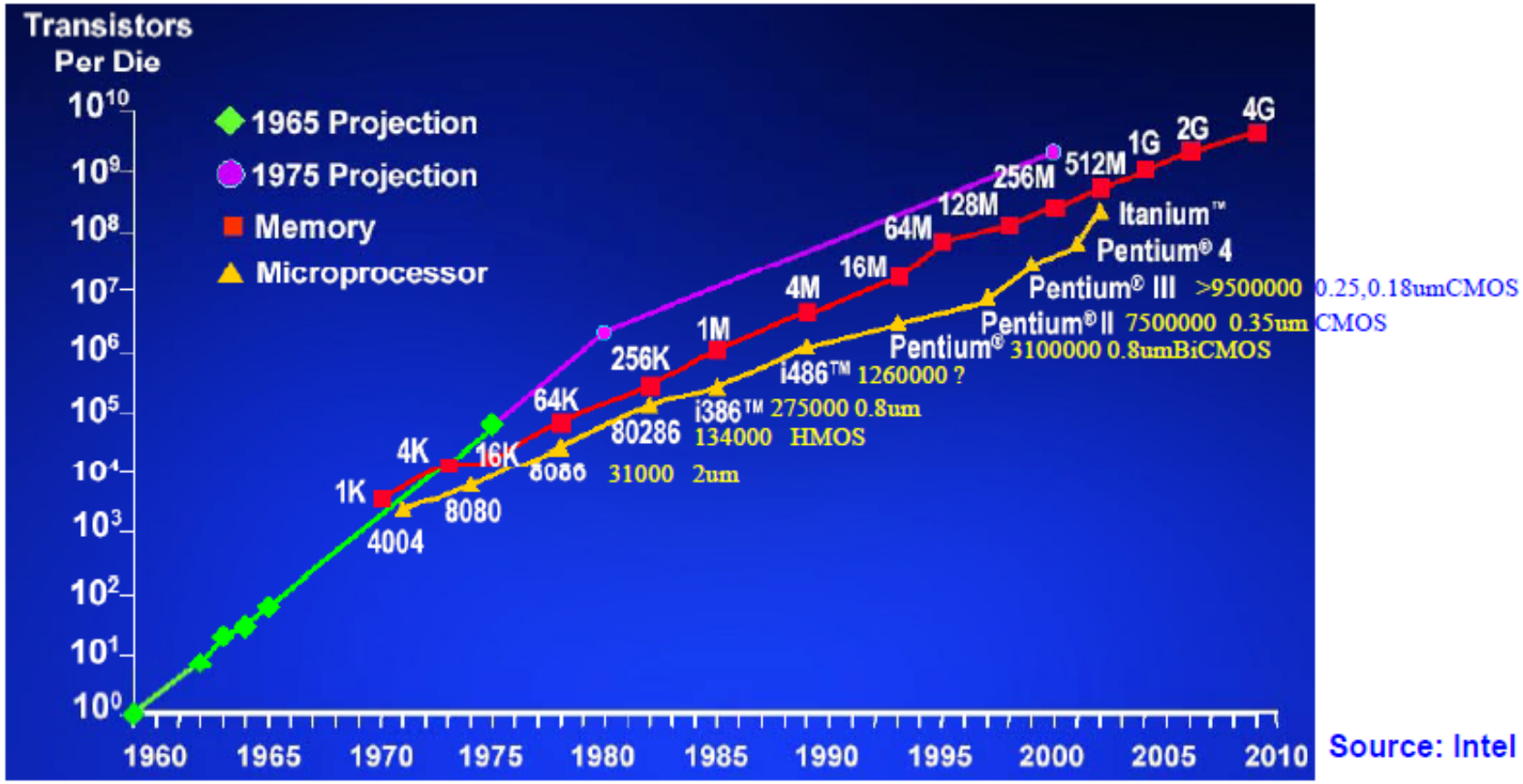
Evolution of Minimum Feature Size



Moore's Law

- Predicted that the number of transistors per chip would grow exponentially (double every 18 months).

Electronics, April 19, 1965.



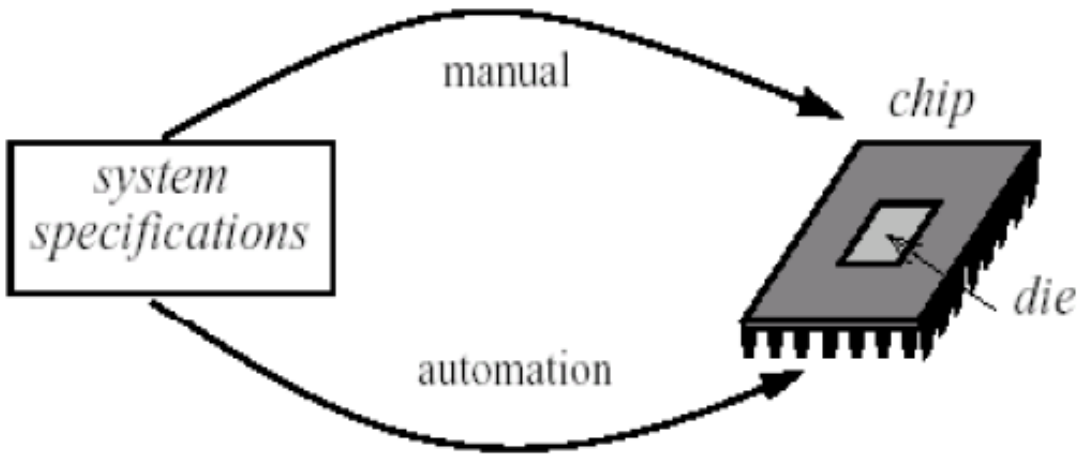
Source: Intel

Semiconductor Technology Roadmap

Year	1997	1999	2002	2005	2008	**2011**	2014
Technology node (nm)	250	180	130	100	70	**50**	35
On-chip local clock (GHz)	0.75	1.25	2.1	3.5	6.0	**10**	16.9
Microprocessor chip size (mm^2)	300	340	430	520	620	**750**	901
Microprocessor transistor/chip	11M	21M	76M	200M	520M	**1.40B**	3.62B
Microprocessor cost/transistor ($x10^{-8}$ USD)	3000	1735	580	255	110	**49**	22
DRAM bits per chip	256M	1G	4G	16G	64G	**256G**	1T
Wiring level	6	6-7	7	7-8	8-9	**9**	10
Supply voltage (V)	1.8-2.5	1.5-1.8	1.2-1.5	0.9-1.2	0.6-0.9	**0.5-0.6**	0.37-0.42
Power (W)	70	90	130	160	170	**175**	183

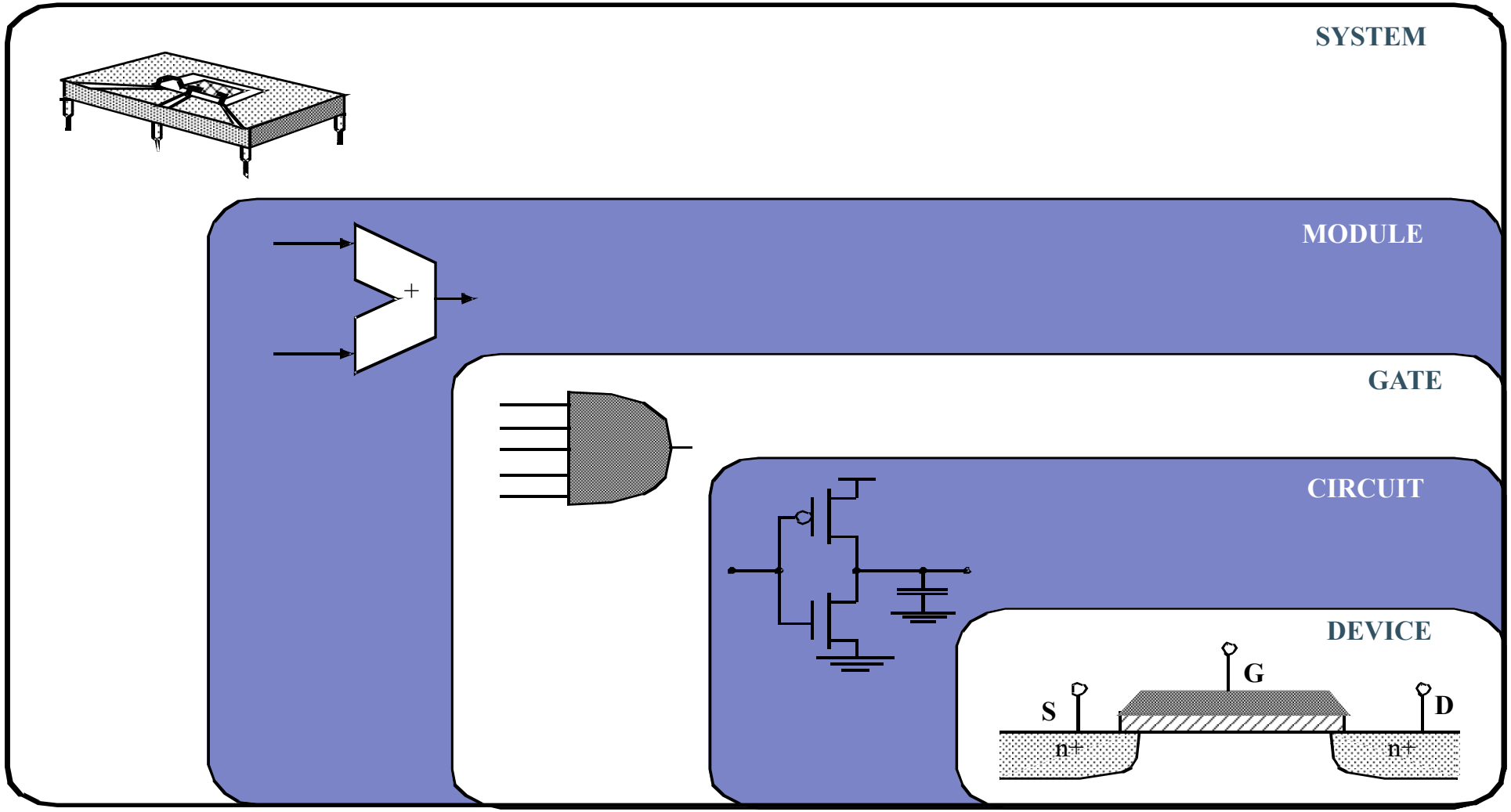
Source: SIA99

VLSI Design Considerations

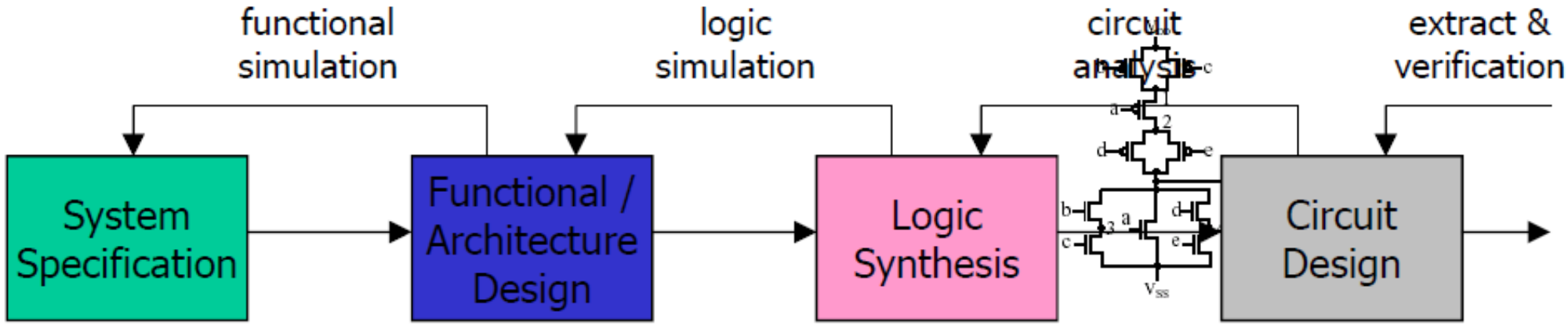


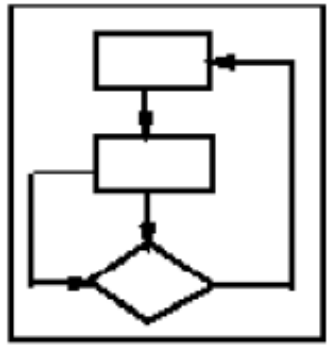
- Several conflicting considerations
  - **Design complexity**: large number of devices/transistors
  - **Cost**: die area, packing, testing, etc.
  - **Performance**: optimization requirements for high performance
  - **Time-to-market**: about 15% gain for early birds
  - Others: power, noise, testability, reliability, manufacturability, etc.

Design Abstraction Levels



Traditional VLSI Design Flow

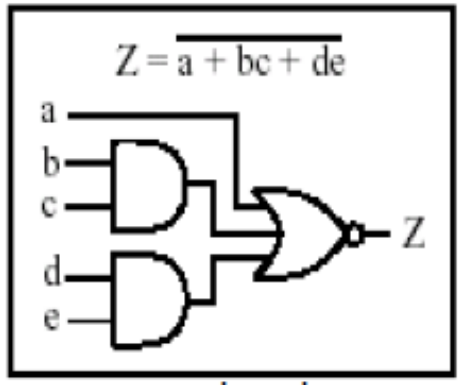




behavior representation

```
module fz(a,b,c,d,e,Z);
input a,b,c,d;
output Z;
assign Z =
  ~(a|(b&C)|(d&e));
endmodule
```

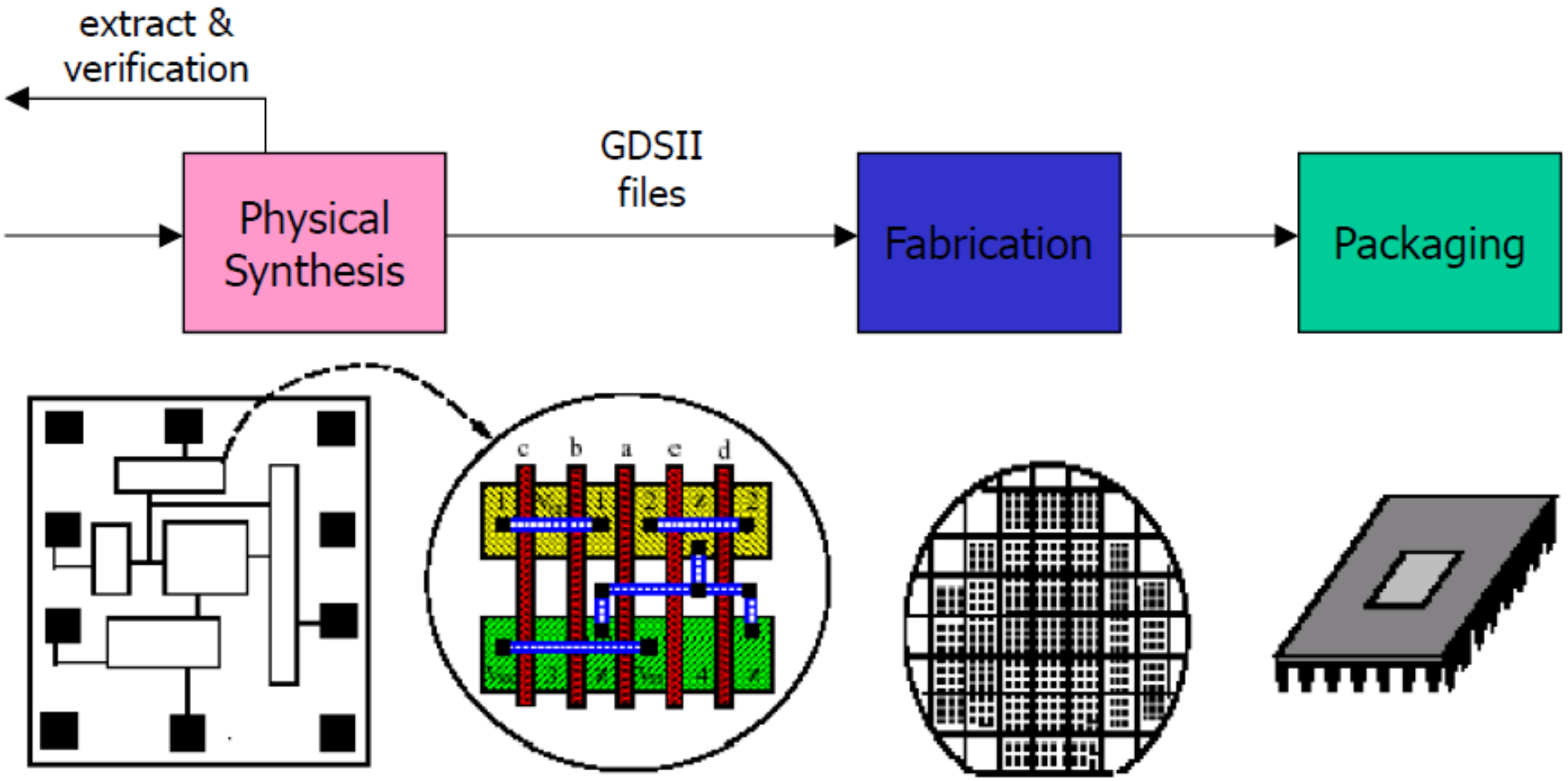
HDL description



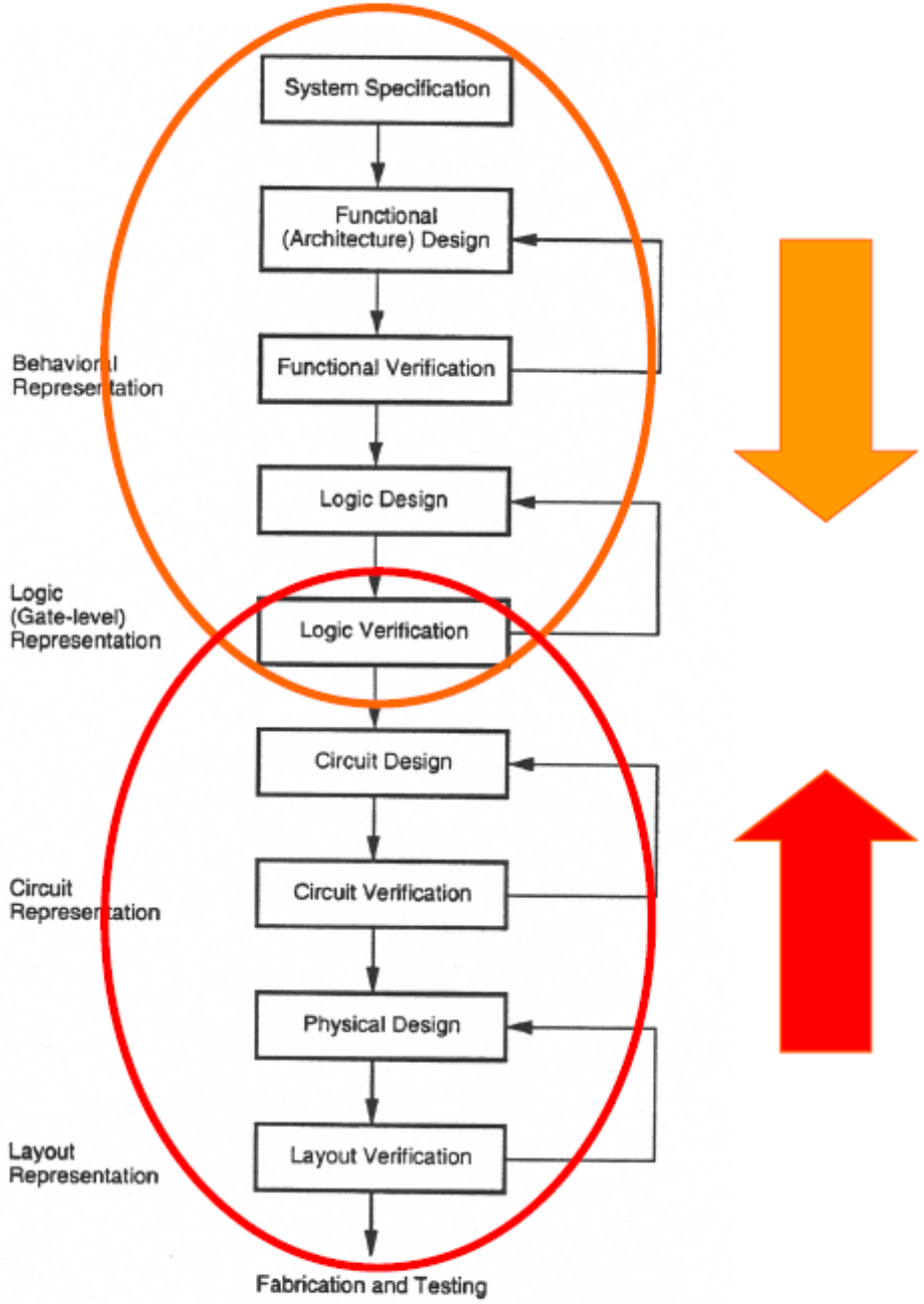
gate-level representation

switch-level representation

Traditional VLSI Design Flow



Simplified VLSI Design Flow



Top-down

Bottom-up

Structured Design Principles

Hierarchy: "Divide and conquer" technique involves dividing a module into sub-modules and then repeating this operation on the sub-modules until the complexity of the smaller parts becomes manageable.

Regularity: The hierarchical decomposition of a large system should result in not only **simple**, but also **similar** blocks, as much as possible. Regularity usually reduces the number of different modules that need to be designed and verified, at all levels of abstraction.

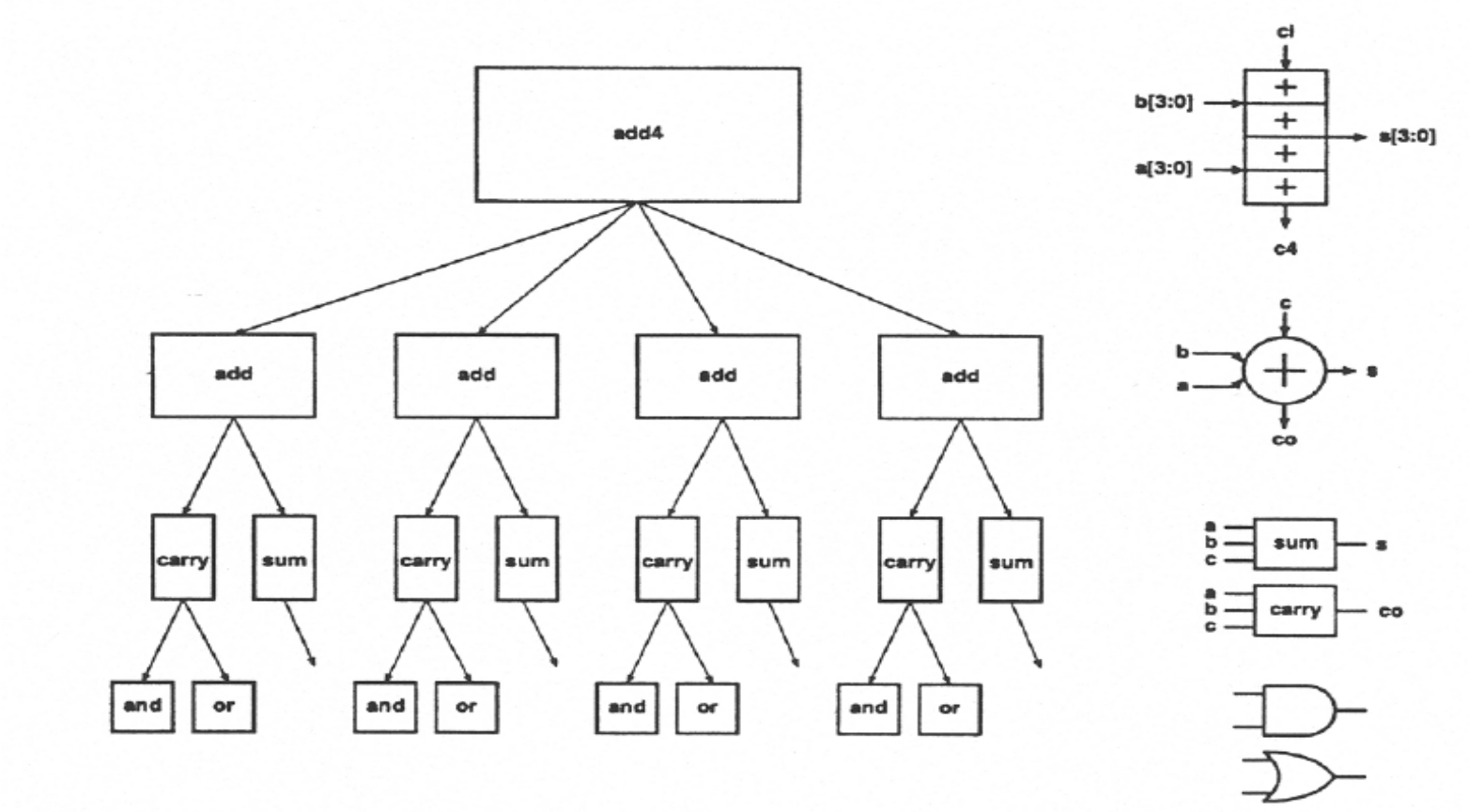
Modularity: The various functional blocks which make up the larger system must have **well-defined functions** and **interfaces**.

Locality: Internal details remain at the local level. The concept of locality also ensures that connections are mostly between neighboring modules, **avoiding long-distance connections** as much as possible.

Design Hierarchy

- The design hierarchy involves the principle of "Divide and Conquer." It is nothing but dividing the task into smaller tasks until it reaches to its simplest level. This process is most suitable because the last evolution of design has become so simple that its manufacturing becomes easier..

Hierarchy of a 4-bit Carry Ripple Adder



Reference

- CMOS digital integrated circuits by Sung –Mo KANG Y.Lebeleci 4th Edition,Mc-Graw Hill publications.